

Marwan Badr

Software Developer | Backend, AI Agent Systems & ERP

marwanbadr514@gmail.com · +20 100 725 8086
github.com/Marwan-badr543 · linkedin.com/in/marwan-badr-4553a42bb
portfolio · Cairo, Egypt



PROFESSIONAL SUMMARY

Software Developer who designs clean, scalable **backend** architectures and secure APIs, specializing in production **AI agents** (multi-agent orchestration, **RAG**, **Human-in-the-Loop**), with **ERP** expertise in Odoo and Frappe / ERPNext and an accounting background.

TECHNICAL SKILLS

Languages	Python, JavaScript, HTML, CSS, SQL
Backend & API	FastAPI, Django, Django REST Framework (DRF), RESTful APIs, Webhooks, AsyncIO
Web Development	Next.js, React, Vite, HTML, CSS, JavaScript, REST & SSE Integration, Responsive UI
Databases & Caching	PostgreSQL, MySQL, MariaDB, Redis, Qdrant (Vector DB)
AI & Agentic Systems	LangGraph, LangChain, Multi-Agent Architecture, Tool Calling, RAG, Structured Outputs (Pydantic), Human-in-the-Loop (HITL), Guardrails, Prompt Engineering
Odoo Development	Odoo 17, Odoo 18, Custom Modules, ORM, XML Views, Security Rules, APIs
Frappe & ERPNext	Frappe Framework (v14, v15), ERPNext, DocTypes, Server Scripts, Client Scripts, Hooks, Jinja
DevOps & Tools	Docker, Git, GitHub, Nginx, Linux / Bash
Desktop Development	PyQt5
Accounting Domain	Professional Financial Accounting (PFA), General Ledger, AR/AP, Bank Reconciliation, Audits
Spoken Languages	Arabic (Native), English (Conversational)

PROFESSIONAL EXPERIENCE

Founder & Lead AI Architect <i>Razyyn AI (Autonomous Accounting Agent SaaS) – Remote</i> • Designed and shipped to production a multi-agent AI accountant (LangGraph, FastAPI) serving ERPNext and Odoo. • Lead backend APIs, native ERP integrations (Frappe, Odoo), and guarded write controls.	<i>Aug 2026 – Present</i>
Software / Frappe Developer <i>Showsatellite (Home Appliances Retail) – Remote (Bahrain)</i> • Own the backend APIs and web platform (customer storefront and admin panel) for the company’s e-commerce operations, integrated with the ERP for orders, stock, and invoicing. • Maintain real-time product, price, and stock synchronization between the ERP and the online store, and the operations panel for delivery scheduling, driver dispatch, technicians, and support tickets. • Lead custom ERPNext development and workflow automation.	<i>Mar 2026 – Present</i>
Accountant <i>Minicash Furniture Company – Onsite</i> • Managed daily accounting cycles, general ledgers, Chart of Accounts, and AR/AP reconciliations.	<i>Feb 2026 – Mar 2026</i>

FEATURED PROJECTS

- Razyyn AI — Autonomous AI Accountant (Founder & Architect):** Production multi-agent SaaS for ERPNext and Odoo – LangGraph router with 6 specialized agents, Qdrant RAG, Pydantic tool calling, guardrails, and HITL-approved ERP writes.
- Razyyn AI — Native Odoo Modules (v17 & v18):** Custom Odoo addons embedding in-ERP chat UI, guarded write gateway, AST-based live row SQL filtering, and multi-company security.
- Razyyn AI — Native Frappe Applications (v14 & v15):** Custom Frappe apps providing ERPNext chat UI, multi-agent modes (Ask, Analyse, Audit), attachment processing, and granular write policy controls.
- Silo — Plugin-Based AI Customer Service Agent:** Open-source LangGraph agent extended through plugins, with RAG over company documents and human-approved ERPNext ticket creation.
- Integrated E-Commerce & Enterprise Operations Platform:** Full-scale customer ordering platform integrated with ERPNext, automating sales workflows, Telegram alerts, and logistics dispatch.
- Real-Time Synchronization Engine: ERPNext ↔ OpenCart:** Bidirectional sync engine automating product catalogs, dynamic pricing, inventory levels, and images in real time with zero variance.
- Office Leaks — Professional Social Network:** Corporate transparency social platform built with Django and DRF, featuring company workplace reviews, culture ratings, and nested discussions.
- Odoo Sales Commission Module:** Custom Odoo 17 module automating sales team commission tracking, computation rules, and payouts linked directly to invoice records.
- Support Management Frappe App:** Service and ticketing app for Frappe/ERPNext managing maintenance schedules, technician visits, and service appointments.

EDUCATION

Menoufia University

Bachelor of Commerce – Accounting (English Section)

Graduated 2025
Grade: Very Good

MILITARY STATUS

Permanently Exempted